

Machine learning: Repetition questions

You should be able to answer all of these questions in the project presentations!

1 Introduction

No repetition questions

2 ML Overview

- What are the two foundations of machine learning?
- Name the 3 main ML problem classes.
- Formally define supervised learning (you may use a whiteboard).
- What is the formal difference between classification and regression?
- Name and formally define the most used error measure for classification.
- Name and formally define the most used error measure for regression.

3 First classifiers and ML Experiments

First classifiers

- What are the resources in an ML project?
- Name the (broadly generalized) steps of an ML project.
- What do you usually try to optimize during an ML evaluation?
- What are the steps of a single ML experiment?
- What is the relation between an ML evaluation and an ML experiment?
- What are general properties of *good* data?
- Name 3 possible classifier categorizations.
- Formally explain the Bayes classifier and write down the formula used.
- Why do we need other classifiers if the Bayes classifier is guaranteed to be optimal?
- Give a (formal or informal) definition of the term "decision boundary".
- How does the kNN classifier work (informal description)?
- How does the training procedure of a kNN classifier work?
- What are the hyperparameters of the kNN classifier?

ML Experiments

- Why do we need a separation between train and test data?
- How is the term "overfitting" defined?
- What do we have to consider when we separate train and test data?
- How can we check that the training data is large enough?
- How can we check if the test data is large enough?
- What is the difference between test and validation data? What do we use these datasets for?
- Give *our* definition of "parameters" and "hyperparameters".
- What are features?
- What is a common trap in feature normalization?
- How is accuracy defined and in which situations can you use it?
- Define Specificity, Sensitivity, Recall, Precision. Be able to do calculations from a 2-class confusion matrix.
- What is an easy to interpret single number quality measure? What is the most meaningful single number quality measure?
- Name at least 2 criticism of the f1 score.
- What is the best/worst value of the auROC?

4 Regression

- Write down the simple linear regression model in a formal way.
- Name the steps that you'll do when computing the linear regression (simple or ordinary) parameter estimation formulas.
- How is the RSE computed?
- How is the R^2 measure computed?
- What advantages does the RSE has compared to the R^2 statistic and vice versa?
- What is the relation between a kNN classifier and a kNN regressor?
- Give at least 2 general rules when to prefer a parametric or a non-parametric model.

5 Basic Classifiers

Logistic Regression

- Give two reasons why we can't use linear regression for classification.
- State the logistic function. How does it look like? Make a sketch.
- There are two ways to extend logistic regression to multiple classes. Which?
- Why does logistic regression has so many function parameters in your favorite ML library?

Bayesian Classifiers

- Write down the Bayes classifier. For which error measure is it optimal?
- Name at least 3 Bayesian type classifiers and the assumptions each of those make.
- How can we easily compute ROC curves for Bayesian type classifiers?

6 Feature Engineering

- Name the two most common methods to transform qualitative data into features.
- What is meant by "feature space lifting"?
- What is an "interaction term"?
- What is the (in terms of result) best feature selection method?
- What statistics can we use for feature selection? Name a typical statistical feature selection method.
- Name and explain at least 2 feature selection methods that are based on classifiers/regressors.

7 Outliers, Cross validation, Resampling

- What should we do when we detect an outlier in our data? Should we remove it?
- What are the advantages and disadvantages of "leave one out cross validation"?
- Describe k-fold cross validation. Why do we use it? What are the dangers of it?
- What is bootstrapping?

8 Optimization

- Explain the iteration scheme of Gradient Descent in one or two sentences.
- What is a black-box optimization problem?
- What is the best optimization method for a small number of possible parameter combinations and a cheap to compute objective function?
- What are the advantages of grid search?
- Explain Monte Carlo search and when we use it.
- Name at least 3 advanced optimization methods.
- When do we usually use Bayesian Optimization?

9 Advanced Classifiers

- Explain the concept of the SVM classifier (perhaps with the example given in the lecture) in a few sentences.
- What does the parameter C control in the SVM?
- What is the idea of the kernel trick and how does it differ from feature lifting?
- What is tree pruning?
- What is the main advantage of the basic decision tree classifier?
- Name at least 4 parameters that influence random forest computation that we talked about in the lecture!